Lecture 12

Orientation, integration and Stokes' theorem

Objectives. *An orientation specifies the sign of a top-degree form. It lets us integrate forms on a manifold. Stokes' theorem identifies the integral of an exterior derivative with the integral on the boundary, using outward-first orientation. We prove it in every dimension and apply it to the flux of the Coulomb form.*

By the end of this lecture you will be able to:

- Determine orientability using charts and top forms.
- Define an integral independent of the oriented charts and partition of unity.
- Prove Stokes' theorem with its support and orientation hypotheses.
- Compute Coulomb flux by removing a ball around its singularity.

Unit 12A Orientation and integration

12.1 Orientations and oriented charts

Definition 12.1 (Orientation of a vector space)**.** For an n-dimensional real vector space with $n \geq 1$, two ordered bases have the same *orientation* if their change-of-basis determinant is positive.

There are two classes.

Definition 12.2 (Smooth frame)**.** A *frame* on an open set is an ordered list of smooth tangent fields that is a basis at every point.

Definition 12.3 (Orientation of a manifold)**.** An *orientation of a manifold* chooses a class in each tangent space, locally represented by a smooth frame. A manifold is *orientable* if it admits such a choice, and *oriented* when a choice has been made.

Definition 12.4 (Oriented atlas)**.** An *oriented atlas* has positive Jacobian determinants on all chart transitions.

Its coordinate frames give compatible orientation classes: the change of frame uses the inverse Jacobian, whose determinant is also positive. Conversely, near any point an orientation is represented by a frame. Its determinant relative to a coordinate frame is continuous and nonzero. Shrink to a connected chart so that the sign is constant. If necessary replace x^1 by $-x^1$. The coordinate frame then represents the chosen orientation. Charts

obtained this way have positive transitions. This proves the equivalence of orientations and oriented atlases.

A nowhere-zero top form μ selects the bases on which its value is positive. These choices are locally represented by coordinate frames, because its coefficient in a connected chart has constant sign.

Definition 12.5 (Orientation-preserving diffeomorphism)**.** For oriented manifolds, a diffeomorphism is *orientation-preserving* when its differential sends positive bases to positive bases.

In positive charts this means its Jacobian determinant is positive.

For example, on $S^n \subset \mathbb{R}^{n+1}$ the form

$$\mu_p(v_1, \dots, v_n) = \det[p\ v_1\ \cdots\ v_n] \qquad (v_j \in T_pS^n)$$

is smooth and nowhere zero: the radial vector p is independent of the n tangent basis vectors. It defines the outward orientation. For $n = 2$ it is the sphere area form of Lecture 10. Lecture 10 computed the north-pole stereographic inverse pullback as $-4\mathrm{d}u\wedge\mathrm{d}v/(1+u^2+v^2)^2$. This chart reverses the outward orientation.

In dimension zero an orientation is a choice of sign $+1$ or -1 at each point. Integration of a function then means its signed sum. This convention includes the endpoints of an oriented interval.

12.2 Partitions of unity and orientability

A form may have support in several overlapping charts. We multiply it by nonnegative smooth functions whose sum is one. Each weighted form then has support inside one chart. A locally finite sum permits this construction on a noncompact manifold. Only finitely many pieces meet a compact support.

Definition 12.6 (Locally finite collection)**.** A collection of sets is *locally finite* if every point has a neighbourhood meeting only finitely many of them.

Definition 12.7 (Partition of unity)**.** A smooth *partition of unity subordinate to an open cover* $\{U_i\}$ consists of nonnegative smooth functions ρ_i such that

$$\sum_i \rho_i = 1, \qquad \operatorname{supp}\rho_i \subset U_i, \qquad \{\operatorname{supp}\rho_i\} \text{ is locally finite.}$$

Several functions may be assigned to the same cover member.

Local finiteness makes each sum finite near any point. It also implies that a compact set meets only finitely many supports: choose finitely many such neighbourhoods covering the compact set.

Theorem 12.8 (Partitions of unity · *sketched; normalization and use proved*)**.** *Every open cover of a smooth manifold admits a subordinate smooth partition of unity.*

Sketch of existence. The second-countability assumption in Lecture 7 is used in this construction. Together with local compactness, it gives a countable exhaustion by compact sets, each lying in the interior of the next. Refining the cover on the successive compact layers gives a locally finite cover by coordinate balls with compact closure, with smaller balls still

covering the manifold. Choose a smooth bump on each ball that is positive on its smaller ball and supported in the larger one. The coordinate construction uses the smooth function $e^{-1/t}$ for $t > 0$, extended by zero for $t \le 0$. The locally finite sum $b = \sum_i b_i$ of these bumps is smooth and strictly positive. Thus $\rho_i = b_i/b$ is smooth and nonnegative, has the required support, and sums to one. The refinement and bump construction are the sketched steps. See Tu, Theorem 13.7 and Appendix C. ∎

Proposition 12.9 (Orientability criteria · *proved given partitions of unity*)**.** *A smooth manifold is orientable if and only if it has an oriented atlas, and if and only if it admits a nowhere-zero smooth top form.*

Proof. The atlas equivalence and the top-form-to-orientation implication were proved above. For the converse, take positively oriented charts and a subordinate partition. Extend $\rho_i \mathrm{d}x_i^1 \wedge \cdots \wedge \mathrm{d}x_i^n$ by zero outside its chart. This is smooth because its support lies inside the chart. Their locally finite sum is smooth. Every summand is nonnegative on a positive tangent basis, and at least one has positive coefficient at every point. The sum therefore never vanishes. In dimension zero the constant function 1 is already a nowhere-zero 0-form, and every discrete manifold has an atlas. ∎

A tangent pair on a Möbius band reverses orientation after one circuit of its central circle. The twisted band in the figure is parametrised by

$$m(t,s) = \big((2 + s\cos(t/2))\cos t, (2 + s\cos(t/2))\sin t, s\sin(t/2)\big), \quad 0 \le t \le 2\pi, \quad |s| \le 1/2.$$

Along the central circle, we follow the pair $(\partial_t m, \partial_s m)$. At the common point $m(0,0) = m(2\pi,0)$, the first vector is unchanged and the second is reversed.

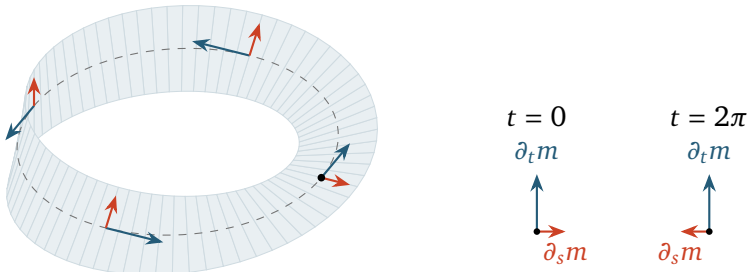


Figure 12.1: The tangent pair follows the central circle of the parametrised band. The two pairs on the right compare its initial and final values at the same point.

The strip model shows the same reversal in quotient coordinates.



Figure 12.2: The red edge arrows mark $(t,s) \mapsto (t+1,-s)$. An upward transverse arrow at $t = 1$ therefore returns at $t = 0$ pointing down.

The open Möbius band is the quotient of $\mathbb{R} \times (-1,1)$ by this identification. On a rectangle of width less than 1, the projection is injective and open: the inverse image of its image is

the open union of all translated copies under the identification. The inverse restrictions give charts on the quotient. Their transitions are restrictions of integer powers of $(t, s) \mapsto (t+1, -s)$, hence smooth. Thus the projection is a local diffeomorphism. If the quotient had a nowhere-zero top form, its pullback would be $h(t, s)\mathrm{d}t \wedge \mathrm{d}s$ with h continuous, nonzero and of constant sign on the connected strip. Invariance under the identification gives $h(t + 1, -s) = -h(t, s)$, contradicting that constant sign. Hence it is not orientable.

For $\mathbb{R}\mathrm{P}^2$, the projection $q\colon S^2 \to \mathbb{R}\mathrm{P}^2$ is a local diffeomorphism by the ratio charts of Lecture 7. On a hemisphere $\sigma X^i > 0$, with $\sigma \in \{+1, -1\}$, its coordinate map is $X \mapsto (X^j/X^i)_{j\neq i}$. Its smooth inverse is $y \mapsto \sigma Y/\|Y\|$, where Y has ith component 1 and remaining components y. This proves the local diffeomorphism assertion. A nonzero top form would pull back to $h\mu$ on S^2, where μ is its outward area form and h has constant nonzero sign. The antipodal map $a(p) = -p$ satisfies

$$(a^*\mu)_p(v, w) = \det[-p\ -v\ -w] = -\mu_p(v, w).$$

Since $q \circ a = q$, invariance forces $h(-p) = -h(p)$, again impossible. Thus $\mathbb{R}\mathrm{P}^2$ is not orientable.

For $n \geq 1$, the ratio charts also make $q\colon S^n \to \mathbb{R}\mathrm{P}^n$ a local diffeomorphism. The same determinant calculation gives

$$a^*\mu = (-1)^{n+1}\mu.$$

For even n, a pulled-back nowhere-zero form would again require a constant-sign function to change sign. For odd n, the outward sphere form is invariant under a. Its local expressions therefore descend through q to a nowhere-zero top form on $\mathbb{R}\mathrm{P}^n$. Hence $\mathbb{R}\mathrm{P}^n$ is orientable exactly for odd $n \geq 1$. In particular, $\mathrm{SO}(3) \cong \mathbb{R}\mathrm{P}^3$ is orientable. This also follows from Lecture 9: left translation carries a basis at the identity to a smooth frame on any Lie group.

12.3 Integrating a top form

Recall the ordinary change-of-variables theorem: under a smooth diffeomorphism $y = f(x)$, integration of a compactly supported function uses the factor $|\det Df|$. In positive coordinates the determinant itself is positive. This is the calculus input to integration of forms.

Definition 12.10 (Integral of a top form)**.** For $n \geq 1$, let ω be a smooth compactly supported n-form on an oriented n-manifold. On a positive chart $x\colon U \to x(U)$ write $(x^{-1})^*\omega = f\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$. If its support lies in U, define its integral to be the ordinary integral of f on $x(U)$. For a general support, choose a positive atlas and a subordinate partition and set

$$\int_M \omega = \sum_i \int_{U_i} \rho_i\omega.$$

Only finitely many terms meet the compact support. Each has compact support inside its chart, so every term is defined.

We construct a partition on $\mathbb{R}$ with its usual orientation. We take $U = (-\infty, 1)$ with coordinate x and $V = (-1, \infty)$ with coordinate $y = 2x$. We define a smooth step using the bump function above:

$$\eta(u) = \begin{cases} e^{-1/u}, & u > 0, \\ 0, & u \leq 0, \end{cases} \qquad a(x) = \frac{\eta(x + 1/2)}{\eta(x + 1/2) + \eta(1/2 - x)}.$$

The denominator is positive everywhere. The step is zero for $x \le -1/2$ and one for $x \ge 1/2$. The functions $\rho_U = 1 - a$ and $\rho_V = a$ form a subordinate partition. Their supports lie in U and V respectively. The figure shows two complementary weights on the bounded interval $[-1, 1]$. Their overlap has total weight one.

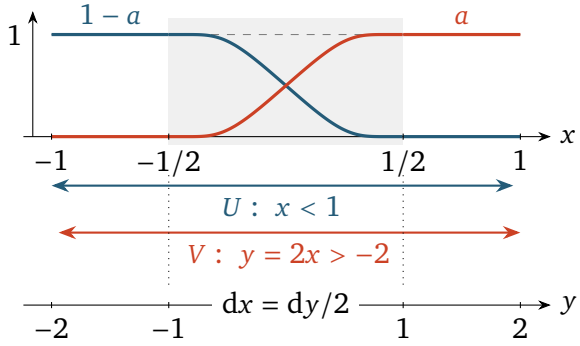


Figure 12.3: The weights $1 - a$ and a sum to one, with $\operatorname{supp} a'$ in the shaded interval. The second coordinate $y = 2x$ doubles lengths, so its form coefficient contains $1/2$.

Example 12.11 (A weighted integral in two charts)**.** We integrate a compactly supported form using these charts and weights. We choose $\omega = a'(x)\mathrm{d}x$, whose support lies in $[-1/2, 1/2]$. In the second chart its expression is $a'(y/2)\mathrm{d}y/2$. The two weighted chart integrals are

$$\int_U \rho_U \omega = \int_{-1/2}^{1/2} (1 - a(x))a'(x)\mathrm{d}x = \left[a - \frac{a^2}{2}\right]_0^1 = \frac{1}{2},$$
$$\int_V \rho_V \omega = \int_{-1}^{1} a(y/2)a'(y/2)\frac{\mathrm{d}y}{2} = \left[\frac{a^2}{2}\right]_0^1 = \frac{1}{2}.$$

The bracket limits are values of the step a. Adding gives $\int_{\mathbb{R}} \omega = 1$, which agrees with $a(+\infty) - a(-\infty) = 1$. The Jacobian factor cancels the coordinate stretching. The weights prevent the overlap from being counted twice.

Lecture 10 used cutoff functions in a sketch. Later lectures extend locally defined fields by zero with them, so we now construct them.

Definition 12.12 (Cutoff function)**.** A smooth *cutoff function* at a point equals 1 near that point and has support inside a chosen neighbourhood.

To construct one, take a chart centred at the point and radii $0 < r_1 < r_2$ such that the closed ball of radius r_2 lies inside the coordinate image of the neighbourhood. Using the smooth step above, set

$$m = \frac{r_1^2 + r_2^2}{2}, \qquad w = r_2^2 - r_1^2, \qquad f(x) = 1 - a\left(\frac{|x|^2 - m}{w}\right).$$

The step's argument is at most $-1/2$ for $|x| \le r_1$ and at least $1/2$ for $|x| \ge r_2$. Hence $f = 1$ on the smaller ball and $f = 0$ outside the larger ball. Extending by zero outside the chart gives the required smooth cutoff on the manifold.

Proposition 12.13 (Independence of choices · *proved given partitions of unity and change of variables*)**.** *The integral is independent of the positive atlas and subordinate partition. Reversing the orientation reverses its sign.*

Proof. On overlapping positive charts, the determinant rule for top-form pullback and ordinary change of variables give the same chart integral. For a smooth coordinate change $y = f(x)$, Lecture 10's wedge convention gives

$$f^*(\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n) = \det(Df)\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

For two partitions ρ_i and χ_j subordinate to positive charts, insert $\sum_j \chi_j = 1$ into each integral for the first partition. The result is the finite double sum of the integrals of $\rho_i\chi_j\omega$. Each term is supported in the chart overlap and has the same value in either chart system. Interchanging the sums and using $\sum_i \rho_i = 1$ gives the second partition's answer. This proves independence. For the opposite orientation, reflect one coordinate. Its derivative has determinant -1, whereas the absolute Jacobian in ordinary integration is 1. The top-form coefficient changes sign, proving the last assertion. ∎

For computations, let $D \subset \mathbb{R}^n$ be a compact parameter region, and let $F\colon D \to M$ be smooth with $F(D) = M$. Here smoothness on D means local smooth extension to open neighbourhoods in $\mathbb{R}^n$. Suppose F restricts to an orientation-preserving diffeomorphism $\operatorname{int} D \to M \setminus E$, where $E = F(\partial D)$. Assume ∂D has Lebesgue measure zero and E has measure zero in every manifold chart. The latter condition is chart-independent by change of variables. Then integration is computed by pullback:

$$\int_M \omega = \int_D F^*\omega.$$

Indeed, insert a subordinate partition ρ_i on M. For each summand, ordinary change of variables on $\operatorname{int} D$ gives the integral of $\rho_i\omega$ on $M \setminus E$. The two null boundaries add zero. Summing the finitely many nonzero chart integrals proves the formula. The angular rectangle for the sphere satisfies these hypotheses. See also Tu, §23.4, Definition 23.10.

Example 12.14 (The angular rectangle). We integrate the outward area form μ on S^2 using

$$D = [0, \pi] \times [0, 2\pi], \qquad F(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

The interior maps diffeomorphically onto the sphere minus the seam $E = F(\partial D)$, a half great circle including the poles. Both null-set hypotheses hold. Lecture 10 gives $F^*\mu = \sin\theta \mathrm{d}\theta \wedge \mathrm{d}\phi$. Since $\sin\theta > 0$ in the interior, the parametrisation preserves orientation. Consequently

$$\int_{S^2} \mu = \int_0^{2\pi}\int_0^{\pi} \sin\theta \,\mathrm{d}\theta\,\mathrm{d}\phi = 2\pi[-\cos\theta]_0^{\pi} = 4\pi.$$

Self-check. Compute the area of the unit sphere using its angular area form. Explain why the negative north-pole stereographic pullback does not make this outward-oriented integral negative.

Answer. The angular integral is $2\pi[-\cos\theta]_0^{\pi} = 4\pi$. The north-pole stereographic chart reverses orientation. We therefore reverse the parameter orientation when using its pullback, which has coefficient $-4/(1+u^2+v^2)^2$ in the standard (u, v) order.

Definition 12.15 (Density). A *density* has local expressions $f|\mathrm{d}x^1 \cdots \mathrm{d}x^n|$ that transform with the absolute Jacobian.

Densities can be integrated without orientation because the ordinary change-of-variables factor has the same law. A top form uses a signed determinant, so its signed integral requires

a chosen orientation. The two transformation laws must be distinguished on nonorientable manifolds. For example, $|\mathrm{d}t\,\mathrm{d}s|$ on the Möbius strip is invariant under $(t, s) \mapsto (t + 1, -s)$, since the absolute determinant is one. It defines an area density on the quotient, although no nowhere-zero top form exists there. Lecture 15 constructs such densities from a metric.

Self-check. State in a sentence what the integral of a compactly supported top form on an oriented manifold is and why a partition of unity is introduced.

Answer. The integral is the sum of the ordinary coordinate integrals of $\rho_i\omega$, whose supports lie inside positive charts. The partition localises the form without counting overlaps twice: $\sum_i \rho_i\omega = \omega$.

Unit 12B Boundary and Stokes

12.4 Manifolds with boundary

Three facts are used below. The boundary is a smooth manifold of dimension $n - 1$. The outward transverse direction does not depend on the boundary chart. Tangent spaces at boundary points have dimension n, and forms can be differentiated there.

The closed disc, the closed ball and the annulus have boundary points with half-space neighbourhoods. Their interior points have ordinary open coordinate neighbourhoods. The figure compares these two types of point on the closed unit disc at 0 and $p = (0, 1)$. Near p, the coordinates $u = x$ and $v = \sqrt{1 - x^2} - y$ put the boundary at $v = 0$ and the interior at $v > 0$.

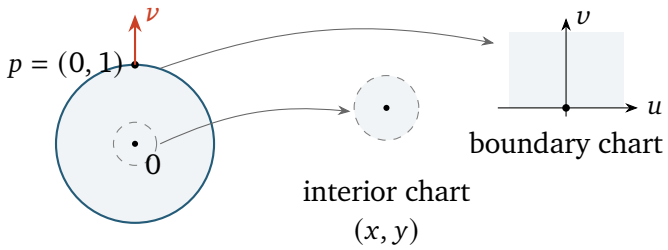


Figure 12.4: An interior point has an open-disc chart. At the top boundary point, the displayed chart sends the disc locally into $v \geq 0$ and the rim into $v = 0$.

The upper and lower closed half-spaces are

$$\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}, \qquad \mathbb{H}^n_- = \{x \in \mathbb{R}^n : x^n \leq 0\}.$$

Definition 12.16 (Manifold with boundary)**.** A smooth *manifold with boundary* is Hausdorff and second countable, with charts onto relatively open subsets of either half-space. Chart transitions and their inverses extend smoothly to neighbourhoods in $\mathbb{R}^n$ near each point.

Definition 12.17 (Boundary and interior)**.** The *boundary* ∂M consists of chart points with $x^n = 0$; the others form the interior.

This geometric boundary is a set of points. Lecture 6's boundary operator acts on chains. When oriented cells are added, their shared faces cancel, leaving the faces along the geometric boundary.

Allowing either half-space permits the coordinate reflection needed for positive charts also in dimension one. At the right endpoint of the oriented interval $[0, 1]$, the positive chart $u = x - 1$ maps into $(-\infty, 0]$. A positive chart increases with x, so it cannot send this right endpoint to zero with its interior on the positive side. In dimensions $n \geq 2$, reflecting a tangential coordinate can correct the orientation while retaining the upper half-space.

Verification of the boundary facts

The set of boundary points is independent of the chart. Indeed, the chain rule on interior points and continuity make transition derivatives invertible also at boundary points. The inverse function theorem then makes their extensions local diffeomorphisms. Such a map cannot send a half-neighbourhood to a full interior neighbourhood: its inverse would have to take points on both sides of the source hyperplane into the half-space. Restricting charts to $x^n = 0$ gives charts of dimension $n - 1$ on ∂M. The restricted transitions make them smoothly compatible.

A transition's last component vanishes on the boundary, so its tangential derivatives vanish there. Its normal derivative is nonzero by invertibility. It sends inward vectors to inward vectors because it sends interior points to interior points. Thus the outward transverse direction is chart-independent, even when the two charts use opposite half-spaces.

At a boundary point, T_pM is the ambient n-dimensional space modeled by a half-space chart, with changes given by the transition Jacobians. Its vectors need not be velocities of two-sided curves remaining in M. In particular the outward vector $-\partial_n$ is allowed in the upper model. Functions and forms are smooth when their chart coefficients extend smoothly to open sets of $\mathbb{R}^n$. Their derivatives at the boundary are independent of extension, since extensions agree on the interior and all their derivatives agree there and extend continuously to the boundary. The same half-chart construction gives tangent changes and exterior differentiation. Partitions of unity and integration extend using restrictions of bumps to half-balls. The preceding normalization and independence proofs then apply.

Definition 12.18 (Induced boundary orientation)**.** On an oriented manifold the *induced boundary orientation* is *outward first*: a boundary basis $(v_1, \dots, v_{n-1})$ is positive if $(\nu, v_1, \dots, v_{n-1})$ is positive in T_pM, where ν is outward transverse.

Outward-first gives the sign-free formula in Theorem 12.20. Changing ν by a tangent vector or multiplying it by a positive scalar does not change this sign. This proves independence of that choice; no metric is required. For the standard orientation on $\mathbb{H}^n$, take $\nu = -\partial_n$. The boundary orientation is represented by

$$\iota_{-\partial_n}(\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n) = (-1)^n \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1} \quad \text{on } x^n = 0.$$

On $\mathbb{H}^n_-$ the outward vector is $+\partial_n$, so the boundary orientation is $(-1)^{n-1}\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^{n-1}$. For $n = 1$ the upper model gives the negative initial endpoint, and the lower model gives the positive final endpoint. The figure compares the upper half-plane with an interval.

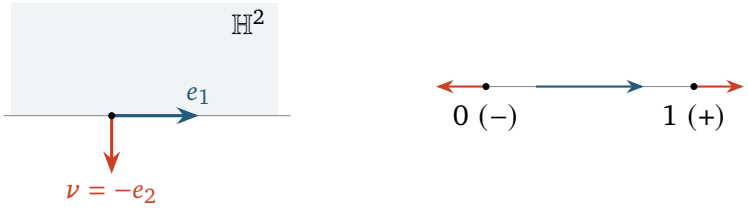


Figure 12.5: On the upper half-plane, the outward vector points down. The boundary tangent points right. On the positively oriented interval, the induced endpoint signs are − and +.

The sphere orientation above is exactly the boundary orientation of the unit ball. The usual Möbius band with its boundary is also nonorientable, since an orientation would restrict to its nonorientable interior.

Definition 12.19 (Closed manifold). A *closed manifold* means a compact manifold without boundary.

12.5 Stokes' theorem in every dimension

For adjacent counter-clockwise squares, a shared edge is traversed in opposite directions. The figure shows this cancellation in a square region with a hole. The remaining outer boundary is counter-clockwise, and the inner boundary is clockwise, as for chains in Lecture 6.

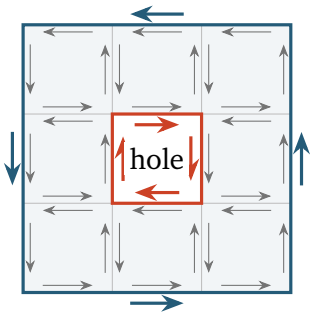


Figure 12.6: Each square has its counter-clockwise boundary orientation. Shared edges cancel in pairs, leaving an outer counter-clockwise boundary and an inner clockwise boundary.

On a coordinate cell the fundamental theorem of calculus gives the integral over its faces. Adding these formulas cancels internal faces. For a general manifold we prove the formula using partitions of unity. The cancellation is expressed by $\sum_i \mathrm{d}\rho_i = 0$.

Theorem 12.20 (Stokes · *proved given partitions of unity*). *Let M be an oriented smooth n-manifold with boundary, where $n \geq 1$. For a smooth compactly supported $(n-1)$-form ω,*

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega.$$

The right side means restriction by the boundary inclusion, with the outward-first orientation. In particular the statement applies to every smooth $(n-1)$-form when M is compact.

Proof. First work on $\mathbb{H}^n$ with compact support. Write the form as

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} f_i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^n, \qquad \mathrm{d}\omega = \Big(\sum_i \partial_i f_i\Big) \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

The derivative formula follows because repeated coordinate factors vanish; the two omission signs cancel for the surviving derivative. Fubini's theorem and the fundamental theorem of calculus integrate $\partial_i f_i$ first in x^i. For $i < n$ the endpoints are outside compact support, so each contribution is zero. The normal contribution is

$$\int_{\mathbb{H}^n} \mathrm{d}\omega = \int_{\mathbb{R}^{n-1}} \int_0^\infty \partial_n f_n(x', x^n)\, \mathrm{d}x^n\, \mathrm{d}x' = -\int_{\mathbb{R}^{n-1}} f_n(x', 0)\, \mathrm{d}x'.$$

On the boundary only the term $i = n$ restricts nontrivially. Its coefficient is $(-1)^{n-1} f_n$. The boundary orientation has the additional sign $(-1)^n$, so its integral is also $-\int f_n(x', 0)\mathrm{d}x'$. For $n = 1$, the same equality uses the negative boundary point and the ordinary endpoint formula. This proves the half-space case in every dimension. On $\mathbb{R}^n$ all coordinate integrations have two endpoints outside support, so the integral of $\mathrm{d}\omega$ is zero, matching its empty boundary. On $\mathbb{H}^n_-$, integrating the normal coordinate from $-\infty$ to 0 gives $+\int_{\mathbb{R}^{n-1}} f_n(x', 0)\mathrm{d}x'$. Its boundary orientation has sign $(-1)^{n-1}$. Multiplying by the same sign in the restricted form gives $+f_n$, so Stokes holds there too. For $n = 1$ this gives the positive final endpoint.

Now choose positively oriented interior and boundary charts on M and a subordinate partition ρ_i. Each $\rho_i \omega$ has compact support inside its chart. In coordinates it extends by zero across the artificial chart edges to a smooth compactly supported form on $\mathbb{R}^n$, $\mathbb{H}^n$ or $\mathbb{H}^n_-$. The local result therefore applies to it. Sum these equalities. Local finiteness and compact support make the sum finite. On M, the product rule gives

$$\sum_i \mathrm{d}(\rho_i \omega) = \Big(\sum_i \mathrm{d}\rho_i\Big) \wedge \omega + \Big(\sum_i \rho_i\Big) \mathrm{d}\omega = \mathrm{d}\omega.$$

Here $\sum_i \rho_i = 1$ and its differential is zero. The restricted partition also sums to one on the boundary. Thus the summed local equalities give the claimed global equality. ∎

The support hypothesis matters on a noncompact manifold. On $\mathbb{R}$, the improper integral of $\mathrm{d}(\tanh x)$ is $1 - (-1) = 2$, while the boundary is empty. The function has no compact support. The nonzero limiting values at infinity remain in the endpoint formula.

12.6 Classical integral theorems and equal fluxes

Definition 12.21 (Compact smooth region)**.** A *compact smooth region* in $\mathbb{R}^n$ means a compact n-manifold with boundary embedded in $\mathbb{R}^n$.

For an oriented manifold S, write $-S$ for the opposite orientation. Stokes includes the integral theorems of vector calculus through Lecture 10's correspondence between vector fields and forms.

- On the interval $[a, b]$ with $a < b$ and increasing orientation, for a smooth function f, the endpoint signs give the fundamental theorem:

$$\int_a^b f'(x)\mathrm{d}x = f(b) - f(a).$$

- On a compact smooth region $D \subset \mathbb{R}^2$ oriented by $\mathrm{d}x \wedge \mathrm{d}y$, for smooth functions P and Q on a neighbourhood of D, Green's theorem is
$$\int_D (\partial_x Q - \partial_y P)\mathrm{d}x \wedge \mathrm{d}y = \int_{\partial D} (P\mathrm{d}x + Q\mathrm{d}y).$$
The outer boundary is counter-clockwise. Boundaries of holes are clockwise.
- On a compact oriented smooth surface $S \subset \mathbb{R}^3$ with smooth boundary, let $\mathbf{A}$ be a smooth vector field on a neighbourhood of S. With the compatible unit normal n, Kelvin–Stokes is
$$\int_S (\operatorname{curl} \mathbf{A}) \cdot n \, \mathrm{d}A = \int_{\partial S} \mathbf{A} \cdot \mathrm{d}\mathbf{x}.$$
Here $\mathrm{d}A$ denotes Euclidean surface area. The positive rim tangent T satisfies that (n, ν, T) is positive in $\mathbb{R}^3$, where ν is the outward tangent normal along the rim. Indeed, (ν, T) is then a positive basis of TS, which is exactly outward-first.

The figure compares these boundary conventions on an interval, a disc and a flat surface in three dimensions.

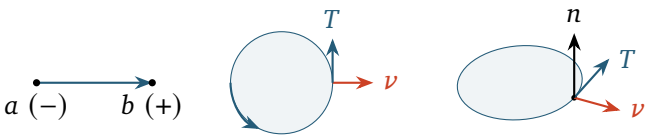


Figure 12.7: Endpoint signs and rim tangents follow the outward-first convention. For the surface panel, the ordered ambient triple is (n, ν, T).

Example 12.22 (Stokes on the unit disc)**.** We take $D = \{x^2 + y^2 \le 1\}$ with orientation $\mathrm{d}x \wedge \mathrm{d}y$ and $\alpha = x\mathrm{d}y$. Its derivative is $\mathrm{d}\alpha = \mathrm{d}x \wedge \mathrm{d}y$, whose disc integral is π by polar coordinates. At $(1, 0)$ the outward vector is e_x, so the positive boundary tangent is e_y. The induced circle runs counter-clockwise. For $c(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, we obtain
$$c^*\alpha = \cos^2 t \mathrm{d}t, \qquad \int_{\partial D} \alpha = \int_0^{2\pi} \cos^2 t \mathrm{d}t = \pi.$$
The boundary and area integrals agree.

Corollary 12.23 (Consequences of Stokes · *proved given partitions of unity*)**.** *For a smooth vector field v on a compact smooth region $K \subset \mathbb{R}^3$, with Euclidean volume $\mu = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$,*
$$\int_K (\operatorname{div} v)\mu = \int_{\partial K} \iota_v \mu.$$
On a closed oriented manifold the integral of an exact top form is zero. If an oriented compact 3-manifold has boundary $S_1 \sqcup (-S_2)$ and a closed 2-form β is smooth on it, then $\int_{S_1} \beta = \int_{S_2} \beta$.

Proof. The first statement follows by applying Stokes to $\iota_v \mu$: Lecture 11 gives $\mathrm{d}\iota_v \mu = (\operatorname{div} v)\mu$. With an outward unit normal ν and a positively oriented orthonormal boundary basis, $\mu(v, \cdot, \cdot)$ is $(v \cdot \nu)$ times $\iota_\nu \mu$, the induced boundary area form. Thus this is the ordinary normal-flux divergence theorem. The empty boundary gives the second statement. Closedness gives $\int \mathrm{d}\beta = 0$; the two induced boundary orientations give $\int_{S_1} \beta - \int_{S_2} \beta = 0$, proving the third. ∎

12.7 Coulomb flux around a missing point

We distinguish the Coulomb form β_C from Lecture 10's ambient form β.

Definition 12.24 (Coulomb flux form)**.** On $\mathbb{R}^3 \setminus \{0\}$ set $r = \sqrt{x^2+y^2+z^2}$ and define the Coulomb flux form by

$$\beta_C = \frac{x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy}{r^3}.$$

The forms β_C and β agree on the unit sphere.

Physical input. In the rationalised electromagnetic units used in Lecture 15, a stationary point charge q at the origin has electric field

$$\mathbf{E}(\mathbf{x}) = \frac{q}{4\pi}\frac{\mathbf{x}}{r^3}, \qquad r = \|\mathbf{x}\| > 0.$$

Under Lecture 10's flux correspondence, β_C is the flux form of $\mathbf{x}/r^3$. Thus the electric flux form is $q\beta_C/(4\pi)$. Closedness of β_C expresses zero divergence away from the charge.

Let K be a compact smooth region with outward-oriented boundary $S = \partial K$ and $0 \notin S$. We compute the outward flux of β_C through S. When the origin lies inside K, a removed small ball has surface S_ε. Its induced inner normal points into the ball, as the section of the region in the figure shows.

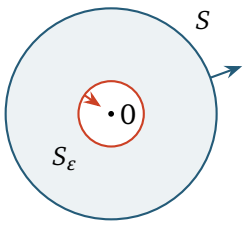


Figure 12.8: The region between S and the small sphere has an inner boundary with normal pointing into the removed ball. Its induced inner orientation is opposite to the ball's outward orientation.

Example 12.25 (Outward Coulomb flux)**.** The derivative of β_C has volume coefficient

$$\partial_x \frac{x}{r^3} + \partial_y \frac{y}{r^3} + \partial_z \frac{z}{r^3} = \frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5} = 0.$$

Thus β_C is closed on its domain. On the unit sphere it is the area form of Lecture 10, so angular coordinates give

$$\int_{S^2} \beta_C = \int_0^{2\pi}\int_0^{\pi} \sin\theta\, d\theta\, d\phi = 4\pi.$$

Here θ is the polar angle and ϕ the azimuth. The coordinate seam and poles have measure zero and do not change this integral. On the sphere of radius $\varepsilon > 0$, radial scaling contributes ε^3 in the numerator and ε^3 in the denominator, so its outward flux is again 4π.

If $0 \notin K$, Stokes applies directly to the closed form on K and gives zero flux. If 0 is inside K, remove an open ball of radius ε whose closure lies in the interior of K. The remaining compact region has boundary $S \sqcup (-S_\varepsilon)$. The form is smooth and closed on this region. The

equal-flux consequence gives $\int_S \beta_C = \int_{S_\varepsilon} \beta_C = 4\pi$. Thus the outward Coulomb flux detects whether the original region contains the origin. For the electric field above, the outward flux is q when 0 lies inside K, and zero when $0 \notin K$. This is Gauss's law for this point charge.

The annulus A in the figure has radii 1 and 2. We write C_s for a circle of radius $s > 0$ with counter-clockwise orientation. The annulus has outward normals pointing away from the region on both circles. The normal at the inner circle points into the hole.

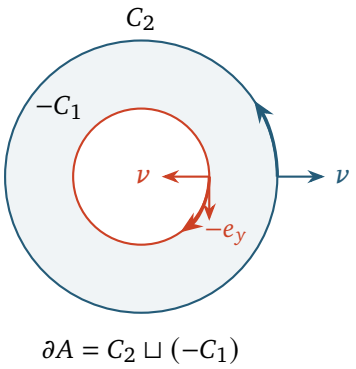


Figure 12.9: The annulus has outer counter-clockwise and inner clockwise boundary orientations. At the inner rightmost point, both the outward normal and positive tangent are negative coordinate directions.

Self-check. Let $A = \{(x, y) : 1 \le x^2 + y^2 \le 4\}$, oriented by $\mathrm{d}x \wedge \mathrm{d}y$, and let

$$\alpha = (x^2 + y^2 + \frac{1}{2})(x\mathrm{d}y - y\mathrm{d}x).$$

Determine both boundary directions and verify Stokes for α on A.

Answer. At $(1, 0)$ the outward vector from A is $-e_x$. The tangent $-e_y$ makes $(-e_x, -e_y)$ positive, so the inner circle is clockwise. The outer circle is counter-clockwise. The derivative and its polar integral are

$$\mathrm{d}\alpha = (4x^2 + 4y^2 + 1)\mathrm{d}x \wedge \mathrm{d}y, \qquad \int_A \mathrm{d}\alpha = 2\pi \int_1^2 (4s^2 + 1)s\mathrm{d}s = 33\pi.$$

For the counter-clockwise circle $c_s(t) = (s\cos t, s\sin t)$ with $0 \le t \le 2\pi$ and $s \in \{1, 2\}$,

$$c_s^*\alpha = (s^4 + s^2/2)\mathrm{d}t, \qquad \int_{\partial A} \alpha = 36\pi - 3\pi = 33\pi.$$

The inner contribution has the minus sign required by its induced orientation.

Summary. Orientations fix signs of top forms. Partitions of unity define a chart-independent integral. Stokes applies to compactly supported forms on oriented manifolds with boundary, using outward-first boundary orientation. Its local proof is the fundamental theorem of calculus in the normal coordinate. Closed forms have equal fluxes across two oriented surfaces that together, with the second orientation reversed, bound a compact region. Removing a ball around the Coulomb singularity gives flux 4π. Lecture 13 studies closed forms modulo exact forms.

Reading. Tu, *An Introduction to Manifolds*, §13 and Appendix C; §§21–23, especially Theorem 23.12, pp. 269–271.